

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Login	Allows users (policymakers, researchers) to create secure accounts and log in to access the application.	Auth Module	Done	0.5
2	HDI Indicator Input Form	A responsive web form where users manually input Life Expectancy, Mean/Expected Years of Schooling, and GNI per capita.	Frontend (Flask UI)	Done	0.5
3	ML Prediction Engine	The core backend logic that processes validated inputs through the trained Random Forest model to predict the HDI score and tier.	Backend (Scikit-learn)	Done	1.5
4	Interactive Data Visualizations	Generates dynamic charts (e.g., bar charts, radar plots) comparing the user's input against global development averages.	Visualization Module	Done	1.0

5	CSV Batch Upload	Allows users to upload a CSV file containing multiple countries' data to get bulk HDI predictions at once.	API / Backend	In Progress	0.5
6	Prediction History & Export	Stores past predictions in the database and allows users to download their results as a CSV or PDF report.	Database / UI	In Progress	0.5
7	Admin Dataset Management	An admin-only interface to upload new annual UNDP data, clean it, and retrain the ML model.	Admin Panel	Pending	0.25
8	Automated Email Reports	Sends the prediction results, tier classification, and visualizations directly to the user's registered email.	API Integration	Pending	0.25

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	4
Core / Must-Have Features	4 (<i>User Auth, Input Form, ML Engine, Visualizations</i>)
Additional / Nice-to-Have Features	4 (<i>Batch Upload, History, Admin Panel, Email Reports</i>)
Features Tested & Verified	4

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	HDI Indicator Input Form, Interactive Data Visualizations
2	Backend / Logic	3	ML Prediction Engine, Data Preprocessing & Validation, Admin Dataset Management
3	Database / Storage	1	Prediction History Storage (SQLite)
4	API / Integration	1	CSV Batch Upload, UNDP/World Bank Data Fetching
5	Security / Authentication	1	User Registration & Login (Password Hashing, Session Management)